SOP-114: Hydrocracker Catalyst Changeout Procedure

Document No: SOP-114-Rev3 | Unit: Hydrocracker-2 (HCU-2) | Effective Date: 2026-03-14 | Review Cycle: 24 months

1. Purpose

This procedure defines the safe sequence of operations for changing out spent hydrocracking catalyst in HCU-2 Reactor R-201, including bed unloading, sulfiding of fresh catalyst, and return-to-service checks. This SOP applies specifically to the NiMo-alumina extrudate catalyst system currently qualified for HCU-2 service.

2. Scope and Applicability

Applies to: HCU-2 Reactor R-201 (Beds 1-3), associated recycle gas compressor K-201, and feed/effluent heat exchanger train E-201A/B. Does not apply to HCU-1 (R-101), which uses a different catalyst loading pattern under SOP-098.

3. Key Process Parameters (Reference Values)

The table below lists the normal operating envelope used to judge catalyst end-of-run:

4. Changeout Sequence

4.1 Unit shutdown and depressurization of R-201 per SOP-050 (Reactor Depressurization).

4.2 Nitrogen purge to < 0.5% hydrocarbon (LEL meter confirmation) before vessel entry permit is issued.

4.3 Spent catalyst unloading via vacuum system, Beds 1-3 in sequence, with continuous H2S monitoring in the confined space.

4.4 Internals inspection: distributor trays, quench zones, thermocouple wells - findings logged on Form F-114-A.

4.5 Fresh catalyst loading (dense loading method) to the bed density specified in the catalyst vendor's Technical Data Sheet, batch-verified against the Certificate of Analysis.

4.6 Catalyst sulfiding (activation) per the sulfiding curve in Appendix A - typical duration 36-48 hours.

4.7 Gradual introduction of feed, WABT ramp not exceeding 15°C/hour until target conversion is reached.

4.8 Post-changeout performance verification against Section 3 parameters for 72 continuous hours before the unit is declared in normal service.

5. Safety Precautions

Pyrophoric catalyst hazard: spent NiMo catalyst can self-heat on air exposure. Unloading must proceed under continuous nitrogen blanket or water-wetting per vendor guidance. Full-face supplied-air respirators are mandatory for all vessel-entry personnel. H2S monitoring is continuous throughout Sections 4.3-4.4.

6. Revision History

This is a synthetic, illustrative document generated for RAG/retrieval testing purposes. It does not represent an actual MRPL internal document.